SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE:CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Annexure A

Debugging & Optimisation Task

Code of Conduct:

I Abarajithan S / 192521299 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S

19/20

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5.A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

		and documentatio n			

1. # IPV4 Network configuration

Given :

-> IP Address : 192.168.10.65

-> Subnet mask : 255.255.255.192 (/26)

-> Default gateway : 192.168.10.1

a) ## Subnet

192.168.10.65 belongs to 192.168.10.64/26

b) ## Network details

-> Network Address : 192.168.10.64

-> Broadcast Address : 192.168.10.127

-> Valid host range : 192.168.10.65 –
192.168.10.126

c) ## verify IP and gateway

-> IP Address (192.168.10.65) - Valid

-> Default gateway (192.168.10.1) - Invalid

d) ## configuration error

-> The default gataway is in different subnet.

e) ## correct configuration

Example :

-> IP Address : 192.168.10.65

-> Subnet mask : 255.255.255.192

→ Default gateway : 192.168.10.126

f) Total usable host

$2^{(32-26)} - 2 = 64 - 2 = 62$

Answer : 62 usable hosts

2. IPv6 configuration

Given :

→ Device A : 2001:db8::ff00:42:832

→ Device B : 2001:db8:0:0:1::1/64

a) Expanded Addresses

Device A : 2001:0db8:000:000:000:ffo
:0042:8329

Device B : 2001:0db8:0000:0000:0001:0
:0000:0001

b) Network prefix

→ Device A : 2001:db8:0:0::/64

→ Device B : 2001:db8:0:0::/64

c) Same subnet ?

→ Yes, both belong to 2001:db8:0:

Address/Prefix mismatch

-> No Prefix mismatch

communication failure may be due to:

-> wrong gateway

-> Firewall

-> interface disable

-> Duplicate address

-> Incorrect routing.

correct configuration

Keep both devices in:

-> 2001:db8:0:0::/64

Example?

-> Device A: 2001:db8:0:0::10/64

-> Device B: 2001:db8:0:0::20/64

Available host Addresses

for /64:

2^{64}

Answer:

-> 18,446,744,073,709,551,616

Addresses (≈ 18 quintillion)

3. VLSM subnet Analysis

Network : 192.168.1.0/24

/26

—> Network : 192.168.1.0

—> Broadcast : 192.168.1.63

—> Host range : 192.168.1.1 - 192.168.1.6

—> Usable host : 62

/27

—> Network : 192.168.1.64

—> Broadcast : 192.168.1.95

—> Host range : 192.168.1.65 - 192.168.1.9

—> Usable host : 30

/28

—> Network : 192.168.1.96

—> Broadcast : 192.168.1.111

—> Host range : 192.168.1.97 - 192.168.1.

—> Usable host : 14

Unused Addresses :

—> /26 —> 64 total, 62 usable

—> /27 —> 32 total, 30 usable

—> /28 —> 16 total, 14 usable

Recommended VLSM

Assign :

-> Largest department -> /26
-> Medium department -> /27
-> Small department -> /28

This minimizes IP wasteges while meeting host requirements.

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4. IPv6 configuration

Expanded addresses

Device A :

2001:0db8:0000:0000:0000:ff00:0042:8329

Device B :

2001:0db8:0000:0000:0001:0000:0000:0001

Network prefix :

Both : 2001:db8:0:0:/64

Same subnet ?

Yes

Possible issues:

→ Gateway

→ Routing

→ Firewall

→ interface configuration

correct example:

→ Device A: 2001:db8:0:0::10/64

→ Device B: 2001:db8:0:0::20/64

Available Address host

2^{64} addresses.

5. Routing table Debugging:

Routing table:

Destination	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default route	10.0.0.1

Destination IP: 192.168.2.100

a) Matching route

matches :

→ 192.168.2.0/24

b) Selected Next Hop

→ 10.0.0.3

c) verify destination

→ 192.168.2.100 belongs to ;

192.168.2.0/24

d) Possible routing errors

→ wrong next-hop IP

→ Missing route on next router

→ Interface down

→ Incorrect subnet mask

→ Firewall blocking traffic.

e) optimized routing :

Ensure :

→ Next hop 10.0.0.3 is reachable.

→ correct subnet mask (/24)

→ interfaces are active.

→ static routes are configured

correctly.

f) If no matching route exists

→ The router uses the default route (0.0.0.0/0)

Next hop: 10.0.0.1.

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